Pranav Rajkumar

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EDUCATION

Bachelor of Science in Aerospace Engineering | Senior Classification at Texas A&M University

PROFESSIONAL SUMMARY

Innovation focused aerospace engineering student with 3 years+ of relevant aerodynamics experience and hands-on composite structures research.

TECHNICAL EXPERIENCE

Carbon Fiber Research Intern

University of Patras | Department of Aeronautical Engineering | Patras, Greece *Summer 2026*

- Developed multi-scale finite element models of carbon fiber reinforced polymer (CFRP) in LS-DYNA, spanning microscale representative volume elements through laminate-level structures
- Extracted full orthotropic elastic constant set ($E11 = 156.1$ GPa) from an RVE loaded in the elastic regime, sourcing periodic boundary conditions for downstream non-elastic load cases
- Built a failure RVE using ADD_EROSION and MAT_089_PLASTICITY_POLYMER keywords to simulate matrix degradation and predict longitudinal strength
- Simulated laminate tension, compression, and interlaminar shear (ILSS) response to generate stress-strain and force-displacement design curves ($E1T = 212$ GPa, $\sigma 1T = 1{,}060$ MPa)
- Applied Mori-Tanaka and inverse rule of mixtures homogenization to map effective modulus and Poisson's ratio across fiber volume fractions

Aerodynamics, Systems, and Controls Engineer

2x World Champion | SAE AERO Design Team | College Station, Texas *05/2024 – present*

- Designed delta wing aircraft producing 37% more lift than previously tested delta & 30% more lift than previous competition design
- Optimized static margin for more payload, increasing mission performance by 20%, breaking school record
- Earned 1st Place Overall, Micro Class at SAE Aero Design East 2026 in Lakeland, Florida
- Utilized Solidworks and Star CCM to optimize control surfaces for all relevant alpha scenarios and flow conditions
- Conducted optimizing trade studies for: airfoil selection, chord length, span length, and vertical stabilizer chord
- Researched and validated motor output for highest static thrust
- Utilized data visualization tools to construct winning design report

Satellite Product Design Team Lead

Aggies Create | College Station, Texas *01/2024 – 05/2024*

- Designed ultra-low cost satellite prototype using off the shelf parts to reduce entry cost for space
- Utilized Solidworks and Onshape to fully 3D print cubesat frame
- Integrated hardware components including gyroscopes, accelerometers, and an IR camera
- Pitched final prototype in front of 10+ investors

Orbital Mechanics and Propulsion Researcher

Human Enabled Venus Design Mission | AIAA | College Station, Texas *09/2023 – 06/2024*

- Designed loitering sub-atmospheric balloon to take advantage of inherent aerial buoyancy of the Venusian atmosphere
- Calculated multiple trajectories for Earth to Venus transit and orbital entry
- Integrated 19+ sensors and instruments in final balloon payload, visualized via CAD modeling
- Interacted and worked with advisors, including Dr. Greg Chamitoff, Space Shuttle and ISS Astronaut

PROJECT EXPERIENCE

Strategic Collections Team Lead

1x Company Record Holder | No Surprise Bill LLC | Denver, CO *06/2025 – present*

- Produced 7+ figures of previously missing revenue for American healthcare providers
- Designed, pitched, and led new talent acquisition and collections strategy to widen revenue funnel for providers
- Broke company record for most revenue located by a collections intern
- Developed new onboarding practices to minimize productivity lost

Tutor

The Tutor Doctor | Houston, Texas *08/2022 – 08/2023*

- Curated specific lesson plans to help individual children learn mathematics
- Taught multiple local and online students mathematics ranging from the elementary to high school level

LEADERSHIP INVOLVEMENT

MSC Freshmen Leaders International

Member | College Station, Texas *09/2023 – present*

- Leadership team member – Speak in front of 60+ people every week, over current events and cultural critique
- Understand the importance of and implement diverse/inclusive spaces on campus
- Learn about and promote interaction with international cultures and students

EXPERT LEVEL SKILLS

SolidWorks (CAD), STAR CCM+ (SIM), LS-DYNA (FEA), Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD), Multi-Scale Composite Modeling, MATLAB, OpenVSP, XFLR5, Wind Tunnel Testing, Project Management, Technical Documentation, Cross-Functional Team Leadership, Strong Analytical Skills